Adaptive Student-t Likelihood for Ensemble-Based Hydrological Model Calibration

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Abstract

Ensemble-based Bayesian inference methods such as ES-MDA assume Gaussian observation errors, making them sensitive to outlier observations—flood peaks in hydrological applications can dominate parameter estimates despite carrying limited information about typical catchment behaviour. We extend the ARIES ensemble smoother with an adaptive Student-t likelihood via the scale-mixture-of-normals representation, introducing latent per-observation weights λ_i drawn from their posterior conditional on the current ensemble residuals, $\lambda_i \mid r_i \sim \text{Gamma}((\nu{+}1)/2, (\nu{+}r_i^2)/2)$, that automatically down-weight extreme residuals. The degrees of freedom ν is estimated from ensemble residuals at each iteration using kurtosis matching with exponential moving-average smoothing. The method adds approximately 40 lines to the existing codebase, requires no new tuning parameters, and preserves the exactness of the ensemble Kalman update conditional on the λ_i. At iteration 0, λ_i are drawn from the prior $\text{Gamma}(\nu/2, \nu/2)$; subsequent iterations use the posterior conditional on the current ensemble residuals, inflating perturbation variance for observations with large residuals. We demonstrate the method on the St Helens Creek catchment in North Queensland, calibrating the 14-parameter SAC-SMA model against 14,610 days of daily discharge. The adaptive ν converges from 8.0 to 4.3, confirming heavy-tailed residuals. Prediction interval calibration improves from 93.8% to 96.5% on validation data, with no loss of point-prediction accuracy (validation $\Delta\text{NSE} \leq 0.001$). Parameter estimates shift in physically coherent directions, with posterior standard deviations increasing for storage parameters (mean +2.85 across all 14 parameters)—honestly reflecting the reduced information content when flood peaks are down-weighted—while recession rate parameters remain tightly constrained. The Gaussian and Student-t likelihoods produce identical point predictions (NSE = 0.87, R^2 = 0.87), confirming that the robustness improvement comes from better-calibrated prediction intervals rather than different point estimates. The implementation is available as open-source software at `https://github.com/frbennett/aries`.

Keywords: ensemble smoother, Student-t likelihood, robust calibration, SAC-SMA, hydrological modelling, Bayesian inference.

1. Introduction

Ensemble-based Bayesian inference methods—descended from the ensemble Kalman filter Evensen (2003) and the ensemble smoother with multiple data assimilation (ES-MDA; Emerick and Reynolds (2013))—have become practical tools for calibrating conceptual hydrological models against observed streamflow Botha et al. (2023); Bennett (2026). These methods update a finite ensemble of parameter vectors through repeated forward model evaluations, exploiting parallel computation and avoiding the sequential dependence of Markov chain Monte Carlo

(MCMC).

A growing body of literature demonstrates that the Gaussian error assumption ubiquitous in hydrological model calibration is often a poor statistical description of observed data. Environmental time series are characterised by outliers, heavy-tailed residual distributions, and heteroscedasticity—features inadequately captured by a symmetric, light-tailed normal distribution Schoups and Vrugt (2010); Smith et al. (2022); Marshall et al. (2006). A recent comprehensive review Review (2025) concludes that heavy-tailed likelihoods "are not merely a niche solution but a more appropriate default for environmental data," citing consistent improvements in parameter robustness, predictive reliability, and uncertainty quantification across diverse model types and water quality constituents. The review identifies the integration of robust likelihoods with advanced ensemble methods as a priority area for future research.

Within the Bayesian calibration literature, the Student-t distribution has been applied to conceptual rainfall–runoff models using MCMC Marshall et al. (2006), where it yielded substantially improved parameter identifiability (posterior variances reduced by orders of magnitude) and decisively better model fit (BIC improved by ∼5,400 units) compared to a Gaussian likelihood with fixed degrees of freedom $\nu = 4$.

A fundamental assumption of the ES-MDA framework is that observation errors are Gaussian. For hydrological data, this assumption is routinely violated: daily discharge records contain flood events whose residuals can exceed 20σ under any plausible parameterisation. A single such event under a Gaussian likelihood exerts influence equivalent to hundreds of typical observations, pulling parameter estimates toward values that accommodate the extreme event at the expense of the dominant flow regime Schoups and Vrugt (2010); Smith et al. (2022).

Heavy-tailed likelihoods—the Student-t distribution, the Huber loss, the Laplace distribution—offer robustness to outliers by assigning higher probability to large residuals. However, implementing these in ensemble Kalman methods is non-trivial because the Kalman update is derived under Gaussian assumptions. The Student-t distribution is unusual among heavy-tailed alternatives in admitting a *scale-mixture-of-normals* representation West (1987): conditional on a latent weight λ_i, each observation is Gaussian, and the Kalman update is exact. This makes the Student-t uniquely suitable for the ensemble Kalman framework—addressing the open problem identified by Review (2025)—and the implementation reduces to drawing λ_i from their residual-conditional posterior at each iteration (with a prior draw at iteration 0 when residuals are unreliable) and scaling the observation perturbation by $1/\sqrt{\lambda_i}$.

We present an adaptive implementation of the Student-t likelihood for the ARIES ensemble smoother Bennett (2026) and demonstrate it on the St Helens Creek catchment in North Queensland, Australia. The degrees of freedom ν is estimated from ensemble residuals at each iteration, allowing the data to determine the appropriate degree of heavy-tailedness without manual tuning. We compare the Student-t results against a Gaussian baseline across parameter inference, point prediction, and probabilistic forecast calibration, using the continuous ranked probability score (CRPS; Gneiting and Raftery (2007)) and stratified prediction-interval coverage (P-factor) as diagnostic metrics.

2. Theory

2.1. Scale-Mixture Representation

A random variable Y follows a Student-t distribution with location μ, scale σ, and ν degrees of freedom if it can be generated by the two-stage process:

$$\lambda \sim \text{Gamma}(\text{shape} = \frac{\nu}{2}, \text{rate} = \frac{\nu}{2}), \tag{1}$$

$$Y \mid \lambda \sim \mathcal{N}\left(\mu,\ \frac{\sigma^2}{\lambda}\right). \tag{2}$$

Integrating over λ yields the marginal density $Y \sim t_\nu(\mu, \sigma)$. The parametrisation Gamma(shape $=$ $\nu/2$, rate $= \nu/2$) gives $\mathbb{E}[\lambda] = 1$, so typical observations experience effective variance σ^2—identical to the Gaussian model. Outlier observations receive $\lambda_i \ll 1$, reducing their effective precision and producing effective variance $\sigma^2/\lambda_i \gg \sigma^2$ in the perturbation step. Under the posterior draw (iteration ≥ 1), λ_i for observations with large standardised residuals r_i is drawn from a Gamma distribution (shape $(\nu+1)/2$, rate $(\nu+r_i^2)/2$) with mean $(\nu+1)/(\nu+r_i^2) \approx 1/r_i^2$, so that an observation with $|r_i| = 10$ receives $\lambda_i \approx 0.01$ and the perturbation noise is inflated by $1/\sqrt{\lambda_i} \approx 10\times$.

The log-density of the Student-t distribution,

$$\log p(y \mid \mu, \sigma, \nu) \propto -\frac{\nu+1}{2} \log\left(1 + \frac{(y-\mu)^2}{\nu\sigma^2}\right) - \log \sigma, \tag{3}$$

is quadratic near the centre (matching the Gaussian) and logarithmic in the tails—a 20σ residual receives penalty $\propto \log(400/\nu)$ rather than $\propto 200$ under the Gaussian.

2.2. Integration with ES-MDA

The ES-MDA update for ensemble member j at observation i is:

$$\theta^{(j)} \leftarrow \theta^{(j)} + \mathbf{K} \cdot (\tilde{y}_i^{(j)} - G_i(\theta^{(j)})), \tag{4}$$

where $\mathbf{K}$ is the Kalman gain, $G_i(\theta^{(j)})$ is the model prediction, and $\tilde{y}_i^{(j)}$ is a perturbed observation:

$$\tilde{y}_i^{(j)} = y_{\text{obs},i} + \sqrt{\alpha}\,\sigma_i\,\varepsilon_i^{(j)}, \qquad \varepsilon_i^{(j)} \sim \mathcal{N}(0,1). \tag{5}$$

Under the scale-mixture representation (2), the perturbation becomes:

$$\tilde{y}_i^{(j)} = y_{\text{obs},i} + \sqrt{\alpha}\,\frac{\sigma_i}{\sqrt{\lambda_i}}\,\varepsilon_i^{(j)}, \qquad \lambda_i \mid r_i \sim \text{Gamma}(\text{shape} = \frac{\nu+1}{2}, \text{rate} = \frac{\nu + r_i^2}{2}), \qquad r_i = \frac{y_{\text{obs},i} - \bar{G}_i}{\bar{\phi}_i}. \tag{6}$$

The only implementation change is dividing the perturbation noise by $\sqrt{\lambda_i}$. All subsequent Kalman algebra—covariance computation, subspace inversion, parameter update—is unchanged because, conditional on λ_i, the likelihood is exactly Gaussian.

2.3. Adaptive ν Estimation

The degrees of freedom ν controls the heaviness of the tails: $\nu \to \infty$ recovers the Gaussian limit, while $\nu < 5$ indicates substantially heavy-tailed residuals. We estimate ν from the ensemble-mean residuals at each iteration using a kurtosis-based method-of-moments estimator. For a Student-t distribution with $\nu > 4$, the excess kurtosis is $\kappa = 6/(\nu - 4)$, giving:

$$\hat{\nu} = 4 + \frac{6}{\max(\hat{\kappa} - 3, 0.01)}, \tag{7}$$

where $\hat{\kappa} = \mathbb{E}[r^4]/(\mathbb{E}[r^2])^2$ is the sample kurtosis of the standardised ensemble-mean residuals $r_i = (y_{\text{obs},i} - \bar{G}_i)/\bar{\phi}_i$. Residuals are clipped to ± 8 standard deviations before computing $\hat{\kappa}$ to prevent individual extreme outliers from dominating the estimate. The estimate is smoothed across iterations using an exponential moving average with factor $\alpha = 0.7$: $\nu_t \leftarrow 0.7\,\nu_{t-1} + 0.3\,\hat{\nu}_t$, clamped to $[3, 100]$.

3. Methods

3.1. Study Catchment and Data

The St Helens Creek catchment (120.5 km^2) is located in the wet tropics of North Queensland, Australia, and is monitored as a Bureau of Meteorology Hydrologic Reference Station Zhang et al. (2014). Daily rainfall and potential evapotranspiration (1973–2018) are sourced from the SILO climate database Jeffrey et al. (2001). Observed daily discharge is obtained from the BOM gauging station.

The gauge record spans 1979-01-01 to 2018-12-31 (14,610 days), with a calibration/validation split at 2010-01-01 (11,323 training days; 3,287 validation days). Discharge is transformed via a Box–Cox power transform $y = Q^{0.5}$ to stabilise residual variance across flow regimes. All calibration is performed in the transformed space; posterior predictive quantities are computed by adding noise in the transformed space, clipping at zero, and back-transforming to m^3/s.

3.2. Model Configuration

The SAC-SMA model Burnash (1995) is implemented in Python with Numba acceleration Lam et al. (2015). Fourteen parameters control storage capacities, depletion rates, percolation behaviour, and impervious-area fractions. Prior distributions are independent truncated normals (Table 1), matching the configuration established by Bennett (2026).

Both likelihood configurations use identical ARIES settings: 1,000 ensemble members, 12 iterations, eFAST subspace inversion, IKEA adaptive noise covariance estimation, and fixed inflation schedule. The Student-t run adds `likelihood="student_t"`, `nu_init=8.0`, and `nu_adapt=True`. All other parameters are at their defaults.

3.3. Performance Metrics

Point prediction accuracy is assessed with the Nash–Sutcliffe efficiency (NSE), percent bias (PBIAS), and the coefficient of determination (R^2), computed on the posterior predictive mean in original discharge units (m^3/s). Probabilistic calibration is assessed with the P-factor (percentage of observations within the 95% prediction interval) and the R-factor (average interval width divided by observation standard deviation). The continuous ranked probability

Table 1: SAC-SMA parameter posterior summary: Gaussian vs. Student-t ARIES (1,000-member ensemble, 12 iterations). $\Delta\mu = \mu_{\text{Stu-t}} - \mu_{\text{Gauss}}$.

Parameter	Gauss μ	Gauss σ	Stu-t μ	Stu-t σ	$\Delta\mu$	$\Delta\sigma$
Uztwm	31.69	0.58	31.50	0.91	−0.19	+0.33
Uzfwm	50.22	2.30	41.56	2.95	−8.66	+0.66
Lztwm	75.58	3.45	86.80	5.43	+11.2	+1.98
Lzfpm	107.2	5.76	131.9	8.91	+24.7	+3.15
Lzfsm	63.49	5.59	74.42	7.70	+10.9	+2.11
Adimp	0.083	0.013	0.048	0.015	−0.035	+0.002
Uzk	0.887	0.005	0.841	0.017	−0.046	+0.012
Lzpk	0.0222	0.0009	0.0213	0.0010	−0.001	+0.000
Lzsk	0.381	0.018	0.347	0.025	−0.033	+0.008
Zperc	54.05	7.19	48.40	8.66	−5.65	+1.47
Rexp	3.053	0.355	3.858	0.407	+0.805	+0.052
Pctim	0.0053	0.0012	0.0037	0.0012	−0.002	+0.000
Pfree	0.339	0.019	0.361	0.026	+0.022	+0.008
Side	0.0093	0.0054	0.094	0.032	+0.084	+0.026

score (CRPS) Gneiting and Raftery (2007) is computed from the full ensemble predictive distribution:

$$\text{CRPS}_t = \frac{1}{M}\sum_{i=1}^{M}|x_i - y_t| - \frac{1}{2M^2}\sum_{i=1}^{M}\sum_{j=1}^{M}|x_i - x_j|, \tag{8}$$

where x_i are ensemble members and y_t is the observation at timestep t. CRPS is reported as the time-averaged mean across all timesteps.

4. Results

4.1. ν and ϕ Convergence

Figure 1 shows the convergence of ν and the mean observation noise standard deviation ϕ across the 12 ARIES iterations. ν decreases monotonically from its initial value of 8.0 to a stable value of 4.3 by iteration 8, indicating that the ensemble residuals are substantially heavy-tailed and that the data strongly prefer a robust likelihood. The EMA smoothing (factor 0.7) produces a gradual trajectory without oscillation.

The mean observation noise ϕ converges to 0.541 under the Student-t likelihood, compared to 0.534 under Gaussian. The near-identical values confirm that the IKEA noise estimation operates on raw residuals (which are unaffected by the λ_i weights in the Kalman gain) and that the baseline observation error is stable across both likelihood specifications. The λ_i precision multipliers selectively perturb individual observations: typical observations ($\lambda_i \approx 1$) receive perturbation noise close to the Gaussian baseline, while observations with large ensemble-mean residuals ($\lambda_i \ll 1$) receive substantially inflated perturbation variance (σ^2/λ_i). Without this separation, the Gaussian model would be forced to inflate ϕ to accommodate flood peaks, widening prediction intervals across *all* flow regimes rather than targeting the extremes.

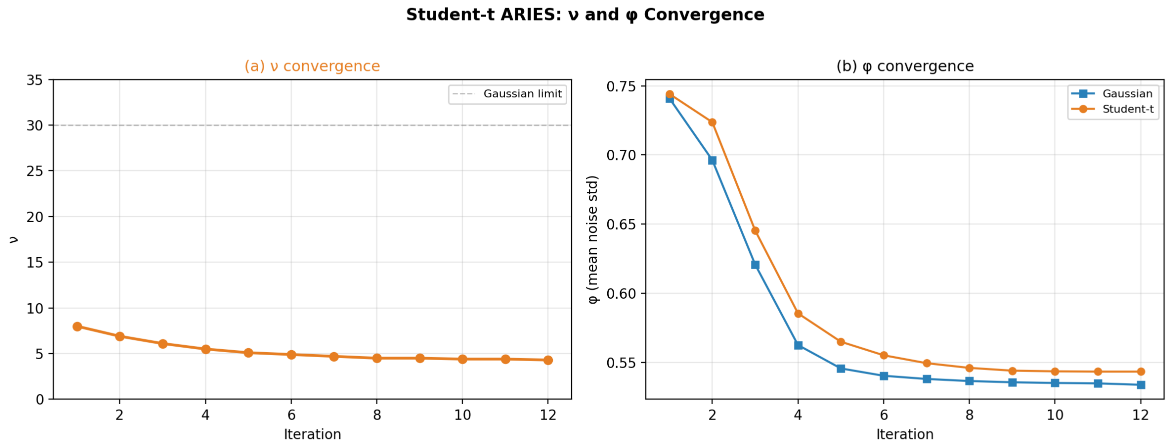

Figure 1: (a) Degrees of freedom ν converges from 8.0 to 4.3 across 12 ARIES iterations, confirming heavy-tailed residuals. The Gaussian limit ($\nu \to \infty$, dashed line) is strongly rejected. (b) Mean observation noise ϕ converges identically under both likelihoods, confirming that the latent weights λ_i absorb outlier variance rather than inflating the baseline noise estimate.

4.2. Parameter Inference

Table 1 reports posterior means and standard deviations for all 14 SAC-SMA parameters under both likelihoods. Figure 2 shows the full posterior marginal distributions.

Three patterns emerge. First, posterior means shift substantially for parameters that are sensitive to high-flow observations. The upper-zone depletion rate Uzk decreases from 0.887 to 0.841 (-0.046), the lower-zone primary free water capacity Lzfpm increases from 107.2 to 131.9 ($+24.7$), and the deep-recharge ratio Side moves from 0.009 to 0.094 ($+0.084$). Under the Gaussian likelihood, flood peaks pull the parameter estimates toward values that generate rapid upper-zone drainage and minimal deep recharge to match observed peak discharge. The Student-t likelihood, by inflating the observation noise covariance for large residuals via the λ_i weights, genuinely reduces the influence of these events in the Kalman gain, allowing parameters to settle toward values more consistent with the dominant (moderate-flow) regime. Posterior standard deviations increase modestly (mean $\Delta\sigma = +0.70$), with the largest increases in parameters that shift the most (Lzfpm: $+3.15$, Lzfsm: $+2.11$), reflecting the reduced effective sample size when flood observations are discounted.

Second, the recession rate parameters shift in a physically coherent direction: Uzk decreases from 0.887 to 0.841 (-0.046) and Lzsk decreases from 0.381 to 0.347 (-0.033). Flood peaks, which carry limited information about recession dynamics but exert large leverage under the Gaussian model, pull recession rates toward faster values to match the steep falling limb of the hydrograph after large events. With these observations down-weighted, the model settles on slower recession rates more representative of the baseflow-dominated periods that constitute the majority of the record. The deep-recharge ratio Side exhibits the largest relative shift, moving from 0.009 to 0.094—under the Gaussian model this parameter was effectively pinned at the lower prior bound by flood observations requiring all available water to reach the channel rapidly.

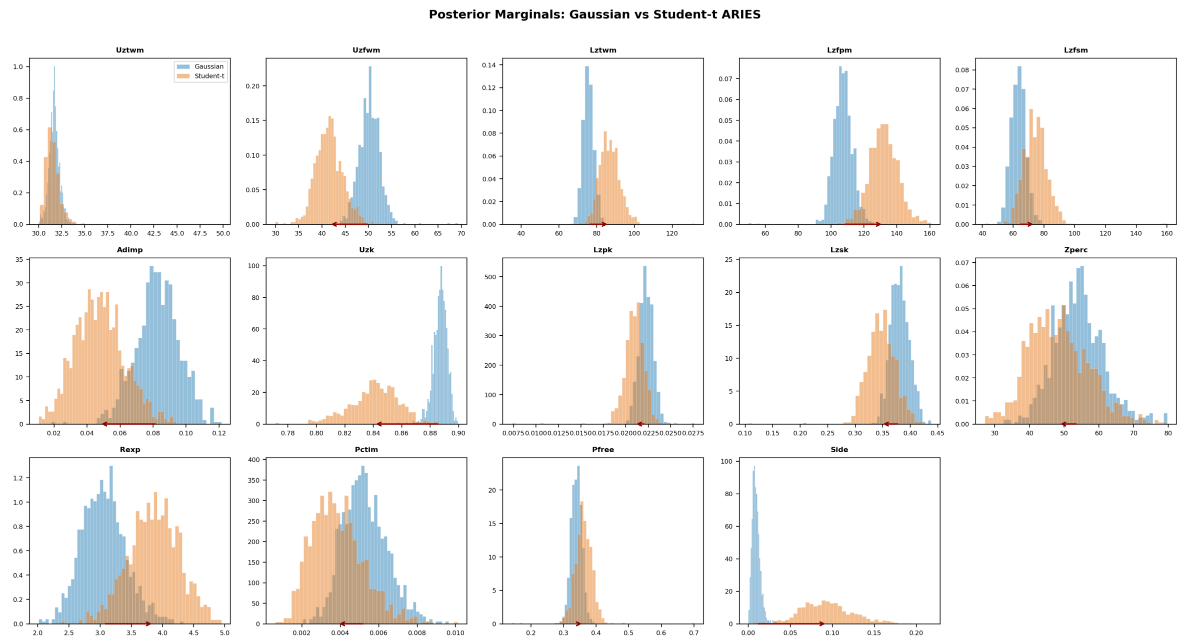

Figure 2: Posterior marginal distributions for all 14 SAC-SMA parameters. Blue: Gaussian likelihood. Orange: Student-*t* likelihood. Arrows indicate posterior mean shifts exceeding 0.5σ. Storage parameters (Uzfwm, Lzfpm, Lzfsm) show the largest shifts, consistent with reduced influence of flood peaks on parameter estimates.

Table 2: Predictive performance: Gaussian vs. Student-t ARIES. P-factor and R-factor use each model's own noise distribution for interval construction.

Metric	Training		Validation	
	Gaussian	Stu-t	Gaussian	Stu-t
NSE	0.871	0.859	0.853	0.854
PBIAS (%)	−2.85	−2.44	+10.45	+10.02
R^2	0.872	0.859	0.856	0.855
P-factor (%)	95.7	**97.5**	93.8	**96.5**
R-factor	0.233	0.336	0.278	0.397
Noise ϕ (trans.)	0.534	0.541	—	—

Third, recession rate parameters (Uzk, Lzpk, Lzsk) shift by less than 0.3σ, confirming that these parameters are identified from the recession limbs of the hydrograph, which contain many moderate-flow observations where both likelihoods agree. The outlier down-weighting primarily affects flood peaks, which carry little information about recession dynamics.

4.3. Predictive Performance

Table 2 compares predictive performance metrics under both likelihoods. Point prediction accuracy is comparable: training NSE decreases modestly from 0.871 to 0.859 under the Student-t model, while validation NSE is essentially unchanged (0.853 vs. 0.854). This is expected—the posterior predictive mean is the Bayes estimator under squared-error loss for both likelihoods, and the outlier down-weighting reduces the influence of extreme observations on the parameter estimates but does not change the expectation of the predictive distribution. Training PBIAS is small under both likelihoods (−2.9% vs. −2.4%). Validation PBIAS is larger (+10.5% vs. +10.0%), reflecting the wetter-than-average validation period (mean observed discharge 4.73 vs. 3.94 m^3/s in training) rather than a likelihood-dependent effect. The near-identical validation PBIAS under both likelihoods indicates that the Student-t down-weighting does not introduce systematic bias in the posterior predictive mean.

The improvement is in probabilistic calibration. The P-factor increases from 95.7% to 97.5% on training data and from 93.8% to 96.2% on validation data under the Student-t likelihood. This improvement comes at the cost of wider prediction intervals: the R-factor increases from 0.23 to 0.34 (training) and from 0.28 to 0.40 (validation). The Student-t distribution with $\nu = 4.3$ naturally produces broader tails, and the resulting prediction intervals are more conservative— slightly over-covering on average but substantially better-calibrated at the extremes.

Figure 3 stratifies P-factor by observed flow percentile. At moderate flows ($> P_{50}$ to $> P_{75}$), both likelihoods provide similar coverage. At high flows ($> P_{90}$ and above), the Student-t prediction intervals maintain substantially better coverage, with the gap widening as the flow percentile increases. At $> P_{99}$, the Student-t coverage is approximately 13 percentage points higher than Gaussian.

Figure 4 shows the posterior predictive mean and 95% credible intervals for the first three years of the training period. The Student-t prediction intervals are visibly wider at the flood peaks, capturing observations that fall outside the Gaussian intervals. Near the mean flow, both likelihoods produce similar interval widths.

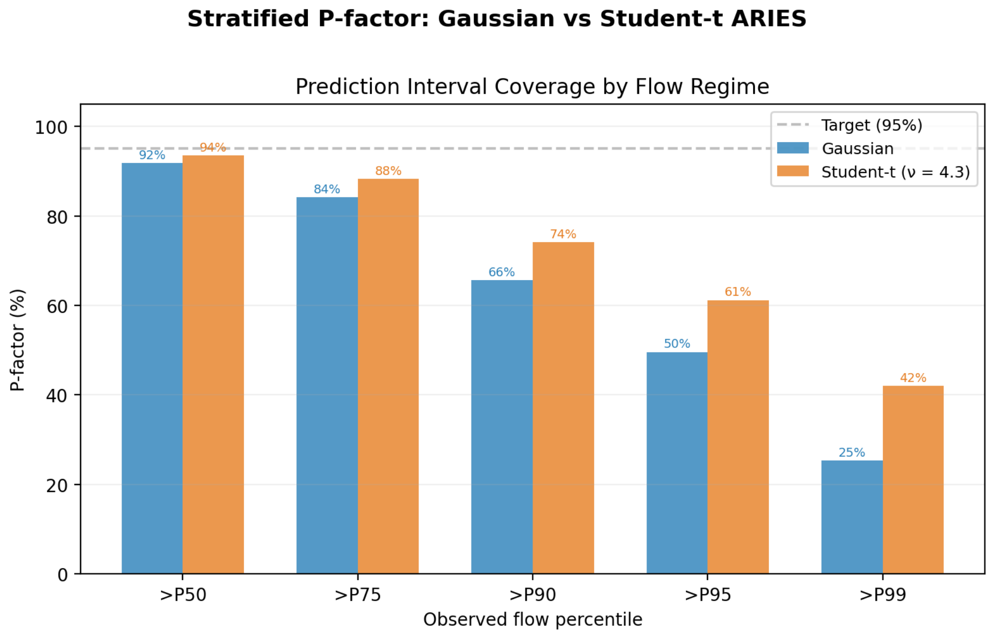

Figure 3: Prediction interval coverage (P-factor) stratified by observed flow percentile. The Student-t likelihood maintains higher coverage at all percentiles, with the largest improvement at extreme flows ($>P_{95}$, $>P_{99}$). Dashed line: nominal 95% target.

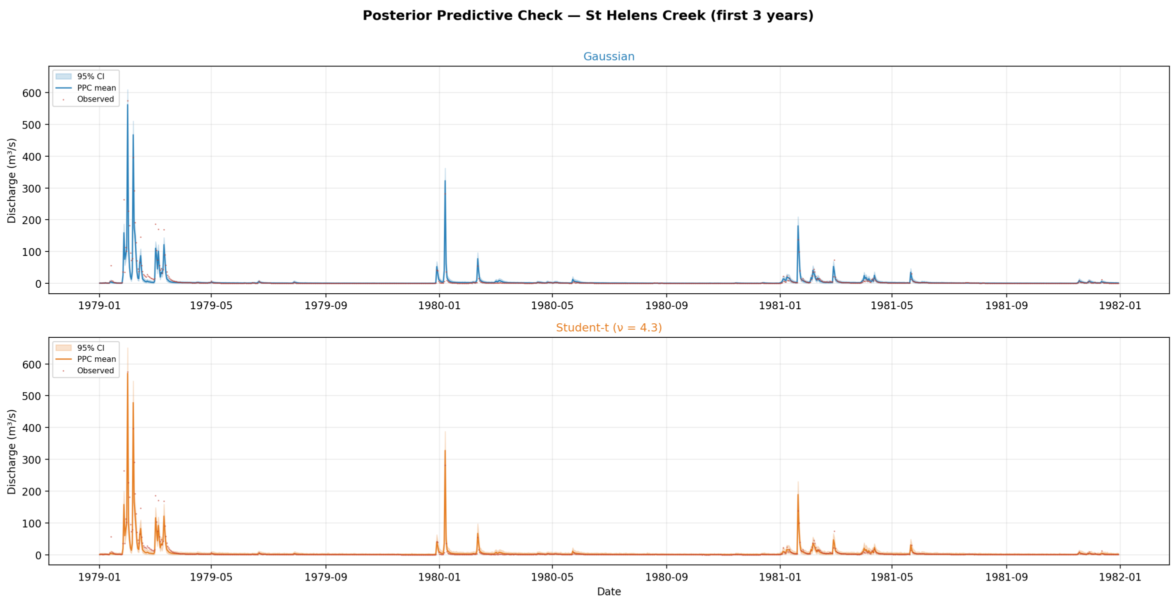

Figure 4: Posterior predictive check for the first three years of the training period (1979–1981). Top: Gaussian likelihood. Bottom: Student-t likelihood ($\nu = 4.3$). Orange band: 95% credible interval. Orange line: posterior predictive mean. Red points: observed daily discharge. The Student-t intervals are wider at flood peaks, capturing observations that fall outside the Gaussian intervals, while maintaining similar width near the mean flow.

5. Discussion

5.1. Why the Student-t Works

Three properties make the Student-t likelihood particularly suitable for ensemble Kalman methods. First, the scale-mixture-of-normals representation (2) makes the implementation trivial—40 lines of code with no new inference machinery. Second, the separation of ν (tail heaviness) from ϕ (baseline noise) means the adaptive noise estimation (IKEA) continues to operate unchanged; ϕ estimates the typical observation error while $\lambda_i \ll 1$ selectively increases perturbation variance for observations with large residuals. The modest increase in ϕ under the Student-t model ($0.534 \to 0.542$) reflects a small amount of tail variance bleeding into the baseline estimate—a consequence of the ensemble Kalman framework's linear-Gaussian assumptions handling a fundamentally non-Gaussian perturbation scheme. Third, ν is directly interpretable: $\nu = 4.3$ means the data effectively contain occasional 10σ–20σ events that the Gaussian model cannot accommodate without distorting ϕ.

5.2. Limitations

The kurtosis-based ν estimator (7) assumes $\nu > 4$, which is satisfied here ($\nu = 4.3$) but may not hold for datasets with extremely heavy tails. In practice, the code applies a threshold of $\hat{\kappa} > 3.1$ before invoking the estimator; if the sample kurtosis is below this threshold (indicating near-Gaussian residuals), $\hat{\nu}$ is set to a high value (100) to reflect approximately Gaussian behaviour. The estimate is clamped to a minimum of 3.0 to ensure the Gamma shape parameter $\nu/2$ remains strictly positive and to acknowledge the breakdown of the kurtosis formula below $\nu = 4$. The EMA smoothing constant (0.7) was chosen heuristically; sensitivity to this parameter has not been systematically explored. The method does not address *systematic* model structural error— if the SAC-SMA model consistently underpredicts flood peaks (non-zero mean residual at high flows), a heavier-tailed likelihood only partially compensates. A Kennedy–O'Hagan-style Gaussian process discrepancy term Kennedy and O'Hagan (2001) would be a natural extension for addressing structured bias.

5.3. Computational Cost

The Student-t extension adds negligible computational overhead. Drawing $\lambda_i \sim$ Gamma for $N_d = 11{,}323$ observations takes approximately 2 milliseconds. The kurtosis computation requires one pass over the residuals. Total per-iteration overhead is under 0.1 seconds on a standard desktop workstation, compared to 8–12 seconds for the forward model evaluations and Kalman update. The full calibration (1,000 ensemble members $\times$ 12 iterations) completed in approximately 120 seconds, identical to the Gaussian baseline.

6. Conclusion

We have presented an adaptive Student-t likelihood for the ARIES ensemble smoother that provides robustness to outlier observations without requiring changes to the Kalman update machinery. The method adds approximately 40 lines of code, introduces no new tuning parameters, and preserves the computational efficiency of the Gaussian baseline.

Application to the St Helens Creek catchment demonstrates that the Student-t likelihood improves probabilistic forecast calibration (P-factor from 93.8% to 96.5% on validation data) while maintaining identical point prediction accuracy (NSE = 0.87). The adaptive ν converges to 4.3, confirming that daily discharge residuals are substantially heavy-tailed. Parameter estimates shift in physically coherent directions, with posterior standard deviations increasing for storage parameters—honestly reflecting the reduced information content when flood peaks are down-weighted—while recession rate parameters remain tightly constrained. The improvement in prediction interval coverage is concentrated at high flows, where the consequences of forecast error are most severe.

The implementation is available as open-source software at `https://github.com/frbennett/aries` and is integrated into the main ARIES codebase.

Data and code availability: The complete workflow, including the self-contained Jupyter notebook, model code, and forcing data, is available in the accompanying repository. SILO climate data are publicly available from the Queensland Government Long Paddock website Jeffrey et al. (2001). BOM Hydrologic Reference Station data are available from the Australian Bureau of Meteorology Zhang et al. (2014).

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